Artificial Intelligence Review (2023) 56:12387–12406
https://doi.org/10.1007/s10462-023-10448-w

Neurosymbolic AI: the 3*rd* wave

Artur d'Avila Garcez[1] · Luís C. Lamb[2]

Published online: 15 March 2023
© The Author(s), under exclusive licence to Springer Nature B.V. 2023

Abstract
Current advances in Artificial Intelligence (AI) and Machine Learning have achieved unprecedented impact across research communities and industry. Nevertheless, concerns around trust, safety, interpretability and accountability of AI were raised by influential thinkers. Many identified the need for well-founded knowledge representation and reasoning to be integrated with deep learning and for sound explainability. Neurosymbolic computing has been an active area of research for many years seeking to bring together robust learning in neural networks with reasoning and explainability by offering symbolic representations for neural models. In this paper, we relate recent and early research in neurosymbolic AI with the objective of identifying the most important ingredients of neurosymbolic AI systems. We focus on research that integrates in a principled way neural network-based learning with symbolic knowledge representation and logical reasoning. Finally, this review identifies promising directions and challenges for the next decade of AI research from the perspective of neurosymbolic computing, commonsense reasoning and causal explanation.

Keywords Neurosymbolic AI · Machine learning · Reasoning · Explainable AI · Deep learning · Trustworthy AI · Cognitive reasoning

1 Introduction

Over the past decade, Artificial Intelligence (in particular, deep learning) has attracted media attention, have become the focus of increasingly large research endeavors and have changed businesses. This led to influential debates on the impact of AI in academia and industry (Marcus 2020). It has been claimed that Deep Learning (DL) caused a paradigm shift not only in AI, but in several Computer Science fields, including speech recognition, computer vision, image understanding, natural language processing (NLP), and

✉ Luís C. Lamb
luislamb@acm.org

Artur d'Avila Garcez
a.garcez@city.ac.uk

[1] Data Science Institute, City, University of London, Northampton Square, London EC1V 0HB, UK

[2] Institute of Informatics, Federal University of Rio Grande do Sul, Av. Bento Gonçalves 9500, Porto Alegre, RS 91501-970, Brazil

machine translation (LeCun et al. 2015). The 2019 and 2020 Montréal AI Debates (Marcus 2020), the AAAI-2021 panel on Neurosymbolic AI, and the AAAI-2020 fireside conversation with Economics Nobel Laureate Daniel Kahneman mediated by Francesca Rossi and including the participation of the 2018 Turing Award winners and DL pioneers Geoffrey Hinton, Yoshua Bengio and Yann LeCun, have all pointed to new perspectives on the future of AI. It has now been argued eloquently that the building of a rich AI system, that is, a semantically sound, explainable and ultimately trustworthy AI system, will require a sound reasoning layer in combination with deep learning.

Parallels have been drawn many times now by AI researchers between Kahneman's research on human reasoning and decision making, reflected in his book "Thinking, Fast and Slow" (Kahneman 2011), and the so-called "AI systems 1 and 2", which would in principle be modelled by deep learning and symbolic reasoning, respectively. In this paper, we seek to place 20 years of research in the area of neurosymbolic AI, known as neural-symbolic integration, in the context of the recent explosion of interest and excitement around the combination of deep learning and symbolic reasoning. We revisit early theoretical results of fundamental relevance to shaping the latest research, and identify bottlenecks and the most promising technical directions for the sound representation of learning and reasoning in neural networks. As well as pointing to the various related and promising techniques within AI, Machine Learning (ML) and Deep Learning, the paper aims to help organise some of the terminology commonly used around AI, ML and DL. This seems important at this exciting time when AI becomes popularized and people from other areas of Computer Science and from other fields altogether turn to AI: psychology, cognitive science, economics, medicine, engineering and neuroscience, to name a few. Throughout the paper, as we make the case for neurosymbolic AI, we will also seek to separate factual information from our opinion. We shall support our case by pointing to important theoretical and empirical results of the past 20 years, but also from our observation of the history of the field: the first wave of AI in the 1980s was symbolic—based on symbolic logic and logic programming, and later Bayesian networks; the second wave of AI in the 2010s was neural (or connectionist), based on deep learning. Having lived through both waves and having seen the contributions and drawbacks of each technology, we argue that the time is right for the third wave of AI: neurosymbolic AI.

In Sect. 2, we summarise the current debate around *neurons vs. symbols* from the perspective of the long-standing challenges of variable grounding and commonsense reasoning. In Sect. 3, we review some of the prominent forms of neural-symbolic integration. In Sect. 4, we seek to address neural-symbolic integration from the perspective of distributed and localist forms of representation. We argue for the importance of a focus on representation based on the assumption that representation precedes learning and reasoning (Valiant 2003).[1] In Sect. 5, we delve deeper into the fundamentals of current neurosymbolic AI methods and systems. In Sect. 6, we identify promising aspects of neurosymbolic AI to address exciting challenges for learning, reasoning and explainability. In Sect. 7, we summarise the paper with a list of ingredients for achieving AI and discuss directions for future research to address the challenges for Artificial Intelligence.

[1] The use of the word *representation* here is intended to be broader than its use in *representation learning* in that the choice of vector space precedes the learning of an embedding.

2 Neurons or symbols: the current debate and research question

Deep learning researchers and AI companies have achieved groundbreaking results in areas such as computer vision, game playing and natural language processing (LeCun et al. 2015). Despite the impressive results, deep learning has been criticised for brittleness (being susceptible to adversarial attacks), lack of explainability (not having a formally defined computational semantics or intuitive explanation, leading to questions around the trustworthiness of AI systems), and lack of parsimony—requiring far too much data, computational power at training time or unacceptable levels of energy consumption (Marcus 2020). Against this backdrop, leading entrepreneurs and scientists such as Bill Gates and the late Stephen Hawking have voiced concerns about AI's accountability, impact on humanity and the future of the planet.[2] Since then, the need for a better understanding of the underlying principles of AI has become generally accepted. A key question is therefore that of *identifying the necessary and sufficient building blocks of AI and how systems that evolve automatically based on ML can be developed and analysed in effective ways that make AI trustworthy and fair*.

Turing award winner Leslie Valiant had pointed out already in 2003 that a key challenge for Computer Science is the principled combination of reasoning and learning, building a rich semantics and robust representation language for intelligent cognitive behavior. *"The aim is to identify a way of looking at and manipulating commonsense knowledge that is consistent with and can support what we consider to be the two most fundamental aspects of intelligent cognitive behavior: the ability to learn from experience and the ability to reason from what has been learned. We are therefore seeking a semantics of knowledge that can computationally support the basic phenomena of intelligent behavior"* (Valiant 2003).

The success of deep learning and its now clear limitations have prompted a heated debate around the value of symbolic AI in contrast with neural computation. A main weakness of DL, as Bengio et al state in a recent article, is that *"current machine learning methods seem weak when they are required to generalize beyond the training distribution, which is what is often needed in practice"* (Bengio et al. 2020). In a series of recent debates on AI, Gary Marcus has argued for DL to be enhanced with symbols (Marcus 2020). Marcus argued eloquently for hybrid AI systems and has sought to define what makes an AI system effectively hybrid. Key to the appreciation of Marcus's argument is an understanding of the representational value of the symbolic manipulation of variables in logic.

Motivated by Valiant's arguments and Marcus's insight, the area of neurosymbolic computing has sought to offer a principled way of studying learning and reasoning by establishing provable correspondences between neural models and logical representations (d'Avila Garcez et al. 2002, 2009, 2015; Lamb et al. 2020), which will be discussed in more detail later in the paper. We contend that what is needed next is research onto the specific methods and techniques which seek to address the issues of representation, robustness and extrapolation. Such methods and techniques are to be drawn from a broad perspective of neurosymbolic AI, symbolic ML and DL, knowledge representation and reasoning, which embraces hybrid systems, to address:

(a) Variable grounding and symbol manipulation Embracing hybrid systems requires the study of how symbols may emerge and become useful in the context of what DL

[2] Several reports on the future of AI and its consequences for society have been published in the mainstream media over the past decade. These include the BBC, NY Times, The Observer and others, see e.g., https://www.bbc.com/news/technology-37713629.

researchers have termed *disentanglement*. Once symbols emerge (which may happen at different levels of abstraction, ideally within a modular network architecture), it may be more productive from a computational perspective to refer to such symbols and manipulate them (i.e., compute with them) symbolically. As an example, suppose that it becomes known that a complex neural network serves the purpose of calculating the sum of two handwritten digits provided as input images to the network. Once the images are recognised as digits by the network, it is probably the case that one would prefer the calculation to be made precise and to extrapolate correctly to any sum of digits. This is easily achieved symbolically. Equally, if it is known that a complex neural network has learned the simple function $f(x) = x$ then one would probably wish to dispense with the network and use its symbolic counterpart instead. The challenge, of course, is in the identification of such a precise symbolic description for the complex network. A symbolic description may be obtained inductively, in which case $f(x) = x$ will be only an approximation that might not apply outside of the data distribution. It may be obtained by inspecting the network or even learned using a simplified setting and fewer parameters. In this case, making the assumption that the symbolic description is correct will produce an exact answer x to any query of the form $f(x)$. Reasoning, in many cases too, is required to be exact and not approximate, although there are cases where approximate or human-like reasoning become more efficient than logical deduction.

(b) Commonsense and combinatorial reasoning A key distinction worth making explicit refers to the difference between commonsense knowledge and expert knowledge. While the former is approximate and difficult to specify, the latter strives to be as precise as possible.[3] We believe that, once equipped with a solid understanding of the value of hybrid systems, variable manipulation and reasoning, the debate will be allowed to progress from the question of *symbols versus neurons* to the more precise research question:

> How to **compute and learn** with symbols, inside or outside of a neural network, and how to **compute and reason** efficiently, in a precise or approximate setting?

Foundational work on neural-symbolic integration and neurosymbolic AI models and systems (d'Avila Garcez et al. 2002, 2009; d'Avila Garcez and Lamb 2003; Hammer and Hitzler 2007), discussed next, will be relevant as we embark in the journey to seek to answer the above research question.

3 Forms of neurosymbolic integration

In d'Avila Garcez et al. (2009), correspondences are proven between various logical/symbolic systems and neural networks. The current limits of neural networks are also evaluated. In a nutshell, neural networks are capable of representing propositional logic, non-monotonic logic programming, propositional modal logic and fragments of first-order logic, but not full first-order or higher-order logic. This limitation has prompted the recent work in areas that combine (symbolic) combinatorial reasoning and deep learning (Prates et al. 2019; Cameron et al. 2020), graph neural networks and neurosymbolic AI, including

[3] Both types of knowledge can exhibit imprecision and may require the modelling of uncertainty, e.g., through the use of probability theory. However, commonsense knowledge is implicit and the conclusions derived from it may need to be retracted frequently in the face of new evidence, whereas expert knowledge is expected to produce over time conclusions with high confidence.

logical neural networks, logic tensor networks and extensions thereof to cater for probabilistic languages and neural probabilistic logic programming (Serafini and d'Avila Garcez 2016; Stehr et al. 2022; Lamb et al. 2020; Prates et al. 2019; Riegel et al. 2020; Manhaeve et al. 2018). Logic Tensor Networks (LTN) (Serafini and d'Avila Garcez 2016) use the language of full first-order logic with deep learning, translating logical statements into the loss function rather than into the network architecture as done before. First-order logic statements are mapped onto differentiable real-valued constraints using a many-valued logic interpretation in the interval [0,1]. The trained network and the logic become communicating modules of a hybrid system, instead of the logic computation being implemented by the network. This distinction between constructing neural and symbolic modules that communicate in various ways and constructing translation algorithms from one representation to another, therefore in a more integrated way, will be explored in what follows and is currently at the centre of the debate on neurosymbolic AI.

Among the recent neurosymbolic systems, one can identify quite a variety in range from integrative to hybrid systems: (Mao et al. 2019) can be seen as a loosely-coupled hybridization where image classification is combined with reasoning from text data (; Manhaeve et al. 2018); offers integration by allowing a node in the probabilistic inference tree of symbolic ML system ProbLog (De Raedt et al. 2007) to be replaced by a neural network; (Serafini and d'Avila Garcez 2016) takes another step towards integration by using a differentiable many-valued logic in the loss function of a neural network; (Minervini et al. 2020; Rocktäschel and Riedel 2017) proposes to perform differentiable unification and theorem proving inside the neural network. This variety might have prompted Henry Kautz to propose a taxonony for neurosymbolic AI as part of a keynote speech at AAAI 2020 (Kautz 2022), which we discuss further in Sect. 6.3. Out of the neurosymbolic methods and systems now available, some more integrated others more loosely-coupled, a common question emerges: **what are the fundamental building blocks, the necessary and sufficient ingredients of neurosymbolic AI?** For example, is the use of an attention layer necessary or can it be replaced by richer structure such as graph networks (Lamb et al. 2020; Hochreiter 2022)? Is the explicit use of probability theory necessary, and in this case inside the network or at the symbolic level, or both? Is there a real computational gain in combinatorial problem solving or theorem proving using neural networks or is this task better left to the devices of a symbolic system (Cameron et al. 2020; Lamb et al. 2020; Serafini and d'Avila Garcez 2016)? One thing is now clearer: there is great practical value in the use of gradient-based learning on distributed representations (LeCun et al. 2015).

To start answering the above questions, in this paper, we seek to bring attention to the relevance of adopting a *distributed or a localist representation*. In a localist representation (Page 2000), the relevant concepts have an associated identifier. This is typically a discrete representation. By contrast, in a distributed representation, concepts are denoted by vectors with continuous values. This raises the issue of which representation, localist or distributed, might be adequate in a situation or more appropriate in general. Symbolic ML, epitomized by decision trees and the learning of logic programs (Lloyd 2003), takes a localist approach while neural networks are distributed, although neural networks can also be localist and learn logic programs (d'Avila Garcez et al. 2009).

With either a localist or a distributed representation, within neurosymbolic AI one may identify *systems that translate and encode symbolic knowledge into the set of weights of a network* (d'Avila Garcez and Zaverucha 1999; d'Avila Garcez et al. 2002, 2009; Hammer and Hitzler 2007; Hu et al. 2016) and *systems that translate and encode symbolic knowledge into the network's loss function* (Serafini and d'Avila Garcez 2016; Marra et al. 2019; Xu et al. 2018). The *neural-symbolic cycle* (d'Avila Garcez et al. 2002) translating

symbolic knowledge into neural networks and vice-versa offers a *compiler* for neural networks, whereby prior knowledge is translated into the network, and a *decompiler* whenever symbolic descriptions are extracted from a trained network. The compiler can either set-up the network's initial weights akin to a one-shot learning algorithm which is guided by knowledge, or define a knowledge-based penalty or constraint which is added to the network's loss function.

A third form of neural-symbolic integration has been proposed in Bengio et al. (2020) which is based on changing the representation of the networks into factor graphs. Although change of representation is a worthwhile endeavor on its own right, helping one understand the strengths and limitations of different models and architectures, this third form of integration creates an intermediate representation with factor graphs in between neural networks and logical representations, when it is known that direct translation between networks and logic is possible. As a result, the use of factor graphs as an intermediate representation is yet to be shown useful, perhaps at facilitating the computational complexity of the translations. Section 6 contains more information about the neurosymbolic AI systems in each of the above categories as part of a note which seeks to clarify the terminology being used in AI, ML and DL.

4 Distributed and localist representation

In order to achieve integration between learning and reasoning, neurosymbolic computing requires a bridge between localist and distributed representations (d'Avila Garcez et al. 2002; Hammer and Hitzler 2007). The success of deep learning indicates that distributed representations with gradient-based methods are more adequate than localist ones for learning and optimization. At the same time, the difficulties of neural networks at extrapolation, explainability and goal-directed reasoning point to the need for a bridge between distributed and localist representations. In neurosymbolic computing, the aim is to represent symbolically the knowledge learned by a neural network. Reasoning takes place either symbolically or within the network in distributed form. Despite their differences, both the symbolic and connectionist paradigms share common characteristics. Change of representation offers a way of making sense of the value of different neural models and architectures with respect to what is a more formal and better understood area of research: symbolic logic (d'Avila Garcez and Lamb 2003; d'Avila Garcez et al. 2009).

Neural network-based learning and inference under uncertainty have been expected to address the brittleness and computational complexity of symbolic systems. Symbolism has been expected to provide additional knowledge in the form of constraints for learning (Gori 2018), which ameliorate neural network's well-known catastrophic forgetting or difficulty with extrapolation in unbounded domains or with out-of-distribution data. The integration of neural models with logic-based symbolism is expected therefore to provide an AI system capable of transfer learning and a bridge between lower-level information processing (for efficient perception and pattern recognition) and higher-level abstract knowledge (for reasoning, explainability, extrapolation and planning).

Suppose that a complex neural network learns a function $f(x)$. Once this function is known, or more precisely a simplified description of $f(x)$ is known, computationally it makes sense to use such a representation, not least for the purpose of extrapolation, as exemplified earlier with the $f(x) = x$ function. One could argue that at this point the neural network has become superfluous. Symbol manipulation (once symbols have been

discovered) is key to further learning at new levels of abstraction. This is exemplified well in Marcus (2020) with the use of the concept of a *container* which may be learned from images.

Among the most promising recent approaches to neural-symbolic integration, so-called embedding techniques seek to transform symbolic representations into vector spaces where reasoning can take place through matrix computations over distance functions (Serafini and d'Avila Garcez 2016; Evans and Grefenstette 2018; Yang et al. 2017). In such systems, learning of an embedding is carried out using backpropagation. Most of the research in this area is focused on the art of representing relational knowledge such as $P(X, Y)$ in a distributed neural network. The logical predicate P relating variables X and Y could be used to denote, for example, the *container* relation between two objects in an image such as a violin and its case, which in turn are described by their embedding. This process is known as relational embedding. For representing more complex logical structures such as first order-logic formulas, e.g., $\forall X, Y, Z : (P(X, Y) \rightarrow Q(Y, Z))$, a system named *Logic Tensor Networks* (LTN) (Serafini and d'Avila Garcez 2016) was proposed by extending *Neural Tensor Networks* (NTN) (Socher et al. 2013), a state-of-the-art relational embedding method. Related ideas are discussed in the context of constraint-based learning and reasoning in Gori (2018) and Marra et al. (2020).

Two powerful concepts of LTN are: (1) the grounding of logical concepts onto tensors with the use of logical statements which act as constraints on the vector space to help learning of an adequate embedding, and (2) the modular and differentiable organisation of knowledge within the neural network which allows querying and interaction with the system. Any user-defined statement in first-order logic can be queried in LTN which checks if that knowledge is satisfied by the trained neural network. With such a tool, a user can decide when to keep using a distributed connectionist representation or switch to a localist symbolic representation. This last aspect brings to the fore the question of the emergence of symbols and their meaning. Recent work using the weak supervision of auto-encoders and ideas borrowed from disentanglement have been showing promise w.r.t. learning of relevant concepts which can be re-used symbolically (Chen and Batmanghelich 2019). Related work seeking to explore the advantages of distributed representations of logic include (Yang et al. 2017), which is based on stochastic logic programs, (Evans and Grefenstette 2018), with a focus on inductive programming, and Minervini et al. (2020), based on differentiable programming and theorem proving. Program synthesis based more generally on probabilistic languages other than logic have also flourished as part of the research area known as neurosymbolic programming (Chaudhuri et al. 2021).

5 Foundations of neurosymbolic computing

In symbolic ML, symbols are manipulated as part of a discrete search for the best representation to solve a classification or regression task. The most well-known form of symbolic ML are decision trees, but richer forms of representation exist, in particular relational representations using first-order logic to denote concepts ranging over variables X, Y, Z... within a (possibly infinite) domain, e.g., $\forall X, Y, Z : grandfather(X, Y) \leftarrow (father(X, Z) \wedge mother(Z, Y))$ (the father of someone's mother is that person's grandfather). Probabilistic extensions of this approach seek to learn probability distributions for such logical rules (or functional programs) as a way of accounting for uncertainty in the training data. Work in these areas is probably best

characterised by research in Statistical Relational Learning (SRL) (Raedt et al. 2016; Richardson and Domingos 2006).

All of the excitement and industrial interest in the past 10 years surrounding AI and Machine Learning, though, have come from an entirely separate type of ML: deep learning. Deep learning uses neural networks and stochastic gradient descent to search through a continuous space, also to solve a given classification or regression task, but creating vector-based, distributed representations, rather than logical or symbolic representations. For this reason, such systems are called sub-symbolic. While it is clear now that AI will not be achieved by building expert systems by hand from scratch, but by learning from large collections of data, one would be misguided to conflate all of machine learning or dismiss the role of symbolic logic, which remains the most powerful and adequate representation for the analysis of computational systems (Fagin et al. 2003). As put simply by Moshe Vardi *"Logic is the Calculus of Computer Science"* and, differently from statistics, machine learning can only exist within the context of a computational system.

Specifically, deep neural networks will require a language for description, as also advocated by Valiant. *Neural network-based AI is distributed and continuous*, deals well with large-scale multimodal noisy perceptual data such as text and audio, handles symbol grounding better than symbolic systems since concepts are grounded on feature vectors, and is by definition a computational model, frequently implemented efficiently using propagation of activation and tensor processing units. *Symbolic AI is generally localist and discrete, capable of sophisticated reasoning*, including temporal, epistemic and nonmonotonic reasoning, planning, extrapolation and reasoning by analogy. Neurosymbolic AI has shown that non-classical logics, in particular many-valued logics, offer an adequate language for describing neural networks (d'Avila Garcez et al. 2009; Serafini and d'Avila Garcez 2016; Riegel et al. 2020). As the field of AI moves towards agreement on the need for combining the strengths of neural and symbolic AI, it can turn next to the question: **What is the best representation for Neurosymbolic AI?**

To answer this question, one should seek to be informed by developments in neurosymbolic computing of the past 20 years, and to evaluate in a precise manner the methods, algorithms and applications of neurosymbolic AI. For instance, it is known that current recurrent neural networks are capable of computing the logical consequences of propositional modal logic programs and other forms of non-classical reasoning and fragment of first-order logic programs (Bader et al. 2007; d'Avila Garcez et al. 2009). Obtaining results for full first-order logic has not been possible thus far, which reinforces John McCarthy's claim that neural networks are in essence systems for propositional logic (McCarthy 1988).

Recent efforts at question answering and language translation as well as protein folding classification have highlighted the importance of hybrid AI systems (d'Avila Garcez et al. 2019; Lamb et al. 2020; Jumper et al. 2020). The ideal type of application for a neurosymbolic system, however, should be that where abstract information is required to be reasoned about at a level of abstraction that goes beyond that what can be perceived from data alone, such as complex concept learning whereby simpler concepts are required to be organised systematically before the definition of a higher-level concept can be learned. Such a conceptual structure requires knowledge which is governed by *general rules and exceptions to the rules*, allowing for sound generalization in the face of uncertainty but also capable of handling specific cases (the many exceptions, which are important for the sake of robustness but not statistically relevant).

6 A note about terminology

In Pearl (2019), Turing award winner Judea Pearl offers a critique of machine learning which, unfortunately, conflates the terms *machine learning* and *deep learning*. Similarly, when Geoffrey Hinton refers to *symbolic AI*, the connotation of the term tends to be that of expert systems dispossessed of any ability to learn. The use of the terminology is in need of clarification. Machine learning is not confined to association rule mining, c.f. the body of work on symbolic ML and relational learning (Raedt et al. 2016). The main differences between symbolic ML and deep learning are (i) the choice of representation - localist logical in symbolic ML rather than distributed in the case of DL, and (ii) the non-use of gradient-based learning algorithms by symbolic ML. Equally, symbolic AI is not just about production rules written by hand. A proper definition of AI concerns knowledge representation and reasoning, autonomous multi-agent systems, planning and argumentation, as well as machine learning. In what follows, we elaborate on the above misunderstandings one at a turn.

6.1 Symbolic ML vs. deep learning

In Pearl (2019), Pearl proposes a hierarchy consisting of three levels: association, intervention and counterfactual reasoning, and claims that ML is only capable of achieving association. A purely symbolic or neurosymbolic ML system should in fact be capable of satisfying the requirements of all three of Pearl's levels, e.g., by mapping the neural networks onto symbolic descriptions. It is fair to say in relation to Pearl's top level in the hierarchy—counterfactual reasoning—that progress has only been made recently within ML and that much research is still needed, although good progress is being made towards the extraction of local, measurable counterfactual explanations from black box ML systems (White and d'Avila Garcez 2019). Once a neural network has been endowed with a symbolic interpretation, one has no reason to doubt the ability of a neural system of asking *what-if* questions. In fact, the very algorithm for extracting symbolic logic descriptions of the form $A \rightarrow B$ from trained neural networks (d'Avila Garcez et al. 2002) uses a form of interrogation of the network akin to the intervention of Bayesian models advocated by Pearl. We argue therefore that a more important question is representational: which representation is most effective, deep networks or Bayesian networks? While attempting to answer this question, as well as considering the demands of the practical applications, it is important to recognise that neural networks offer a concrete model of computation trained by differentiable learning algorithms, one which can be implemented efficiently by message passing or propagation of activation, differently from Bayesian networks. A limitation of having such a concrete computational model may be in the difficulty of pure neural networks at modelling rich forms of abstraction which are not dependent on the data (images, audio, etc.) but which exist instead at a higher conceptual level.

6.2 Knowledge representation and reasoning in AI

Complex problem solving using AI requires a much richer language than that of expert systems as suggested by Hinton. AI requires a language that can go well beyond Horn clauses to include relational knowledge, time and other modalities, negation by failure, variable substitution and quantification, etc. The use of first-order logic (FOL) may not require instantiating (or grounding) all possible combinations of the values of the variables (e.g.,

X and Y in a relation $R(X, Y)$), even though most SRL approaches based on FOL do require grounding, making inference time consuming, e.g., Markov Logic Networks (Richardson and Domingos 2006). In *relational reasoning* with neural networks, borrowing from the field of relational databases, it is typically the grounded (and therefore propositional rather than first-order) representation that is learned and reasoned about. To avoid confusion, we would term this latter task *relationship learning*. Two other equally important attributes of a rich language for complex problem solving are *compositionality*, in the sense of the compositionality of the semantics of a logical language, and modularity. It is worth noting that in the original paper about deep learning (Hinton et al. 2006), before much of the attention turned to Convolutional Neural Networks, modularity was a main objective of the proposed semi-supervised greedy learning of stacks of restricted Boltzmann machines (RBMs). The recently-proposed stacked capsule autoencoders (Kosiorek et al. 2019) and neurosymbolic approaches such as Logic Tensor Networks (Serafini and d'Avila Garcez 2016) as well as other semi-supervised or weakly-supervised approaches revive the important stance of modularity in neural computation. Earlier efforts towards modularity in neurosymbolic AI can be traced back to the system for Connectionist Modal and Intuitionistic Logics (d'Avila Garcez et al. 2009). Modal logics with a possible-world semantics have been shown to offer a natural approach to modularity in neural computation when applied to well-known reasoning tasks and puzzles such as the muddy children puzzle (d'Avila Garcez et al. 2009).

We therefore take the view that AI encompasses ML and more. We argue for the combination of statistical ML, knowledge representation (KR), and logical reasoning. By *logical reasoning*, we mean not only classical logic reasoning with the traditional true-false interpretation, but non-classical reasoning including nonmonotonic, modal and many-valued logics (d'Avila Garcez et al. 2009).

More researchers than ever on both sides of the connectionist-symbolic AI divide are now open to studying and learning about each others' tools and techniques. This was not the case until recently. The use of different terminology alongside some preconceived opinion or perhaps idleness, fueled by the way that science normally rewards research carried out in silos, have prevented earlier progress. Synergies will lead to faster progress in AI. It is reassuring to see it happening in this way: the neural information processing community have shown the value of neural computation in practice, which has attracted the curiosity of great minds from symbolic AI. We hope that further collaboration in neurosymbolic AI will help solve many of the outstanding challenges.

6.3 A taxonomy for neurosymbolic AI

Understanding the role of distributed and localist approaches allow us to present an analysis of Henry Kautz's taxonomy for neurosymbolic AI (Kautz 2022). In Kautz's taxonomy, a Type 1 neural-symbolic integration is *standard deep learning*, which is included to note that the input and output of a neural network can be made of symbols, e.g., text in the case of language translation. Type 2 are hybrid systems such as DeepMind's AlphaGo and other *systems where the core neural network is loosely-coupled with a symbolic problem solver* such as Monte Carlo tree search. Type 3 are systems whereby a *neural network focusing on one task (e.g., object detection) interacts via its input and output with a symbolic system* specialising in a complementary task (e.g., query answering). Examples include the neurosymbolic concept learner (Mao et al. 2019) and deepProbLog (Manhaeve et al. 2018). In a Type 4 neurosymbolic system, *symbolic knowledge is compiled into the training set of a*

neural network e.g., Arabshahi et al. (2019); Lample and Charton (2020); Type 4 includes tightly-coupled, localist systems where forms of symbolic knowledge are translated into the initial architecture and set of weights of a neural network, in some cases with guarantees of correctness (d'Avila Garcez et al. 2009), e.g., Logical Neural Networks which create a 1-to-1 correspondence between neurons and the elements of logical formulas (Riegel et al. 2020). Type 5 are tightly-coupled, but distributed neural-symbolic systems where a *symbolic logic rule is mapped onto an embedding which acts as a soft-constraint (a regularizer) on the network's loss function*. These include LTN (Serafini and d'Avila Garcez 2016; Stehr et al. 2022) and Tensor Product Representations (Huang et al. 2017), referred to in d'Avila Garcez et al. (2019) as *tensorization* methods. Finally, Type 6 systems should be capable of *true symbolic reasoning inside a neural engine*, being considered fully-integrated systems. Early work in neurosymbolic AI has achieved this, see (d'Avila Garcez et al. 2009). Some Type 4 systems are also capable of it, but using localist representations and simpler forms of embedding than Type 5. Kautz adds that Type 6 systems should be capable of *combinatorial reasoning*, possibly by using an attention schema to achieve it. Recent Type 6 efforts include (Cameron et al. 2020; Lamb et al. 2020; Prates et al. 2019), although fully-fledged Type 6 systems for combinatorial reasoning do not exist yet.

Research into Type 5 systems will likely focus on the provision of rich embeddings and the study of the *extent to which such embeddings may correspond either to pre-defined prior knowledge or to learned attention mechanisms*. Further research on Type 6 systems will be highly relevant to the *theory of neurosymbolic computing*. In practical terms, a tension exists between effective learning and sound reasoning, which may prescribe the use of a more hybrid approach of Type 3 to 5, or variations thereof such as the use of attention with tensorization. Orthogonal to the above taxonomy, but mostly associated thus far with Type 4, is the study of the limits of reasoning within neural networks. Recently, this has been the focus of analyses of DL in symbolic domains (Tavares et al. 2020), which in our view should include the study of first-order and non-classical logics.

6.4 A note on explainable AI (XAI)

Knowledge extraction is an integral part of neurosymbolic AI. One main difficulty in XAI is the efficient extraction of compact but correct and complete knowledge. It can be argued that a large knowledge-base is not more explainable than a large neural network. Although this may be true at the level of the entire model explainability, in the case of local explanations, i.e., explanations of individual cases, a knowledge-base is certainly more explainable than a neural network, since it offers a trace (a proof history) showing why an outcome was obtained, as opposed to simply showing how propagation of activation through the network has led to that outcome. It is a main goal of knowledge extraction algorithms to seek to derive compact relevant representations from complex networks.

Early efforts at knowledge extraction in neurosymbolic AI as a form of explanation were always evaluated w.r.t. fidelity: a measure of the accuracy of the extracted knowledge in relation to the neural network rather than the data. High fidelity is therefore fundamental whenever XAI is to be claimed to offer a good explanation for a complex network. Unfortunately, many recent XAI methods have abandoned fidelity as a measure of the quality of an explanation, making it easier for an apparently excellent explanation to be wrong, in that it may simply not be a valid explanation of the given ML model. Even better than fidelity, if an XAI method can be shown to be sound (d'Avila Garcez et al. 2001) then it will provably converge to high-fidelity. Knowledge extraction should also allow communication

between users and the ML system. It should offer a way of identifying and correcting for bias in the system, which is a serious and present problem (Kleinberg et al. 2020). As a result of the General Data Protection Regulation, many companies have decided as a precaution to remove protected variables such as gender from their ML system. It is well known, however, that proxies exist in the data which will continue to bias the outcome so that the removal of such variables may serve only to hide bias that otherwise could have been revealed via knowledge extraction (d'Avila Garcez et al. 2001).

In summary, at least two options exist for neurosymbolic AI. In Option 1, symbols are translated into a neural network and one seeks to perform reasoning within the network. In Option 2, a more hybrid approach is taken whereby the network interacts with a symbolic system for reasoning. A third option, which would not require a neurosymbolic approach, exists when expert knowledge is made available, rather than learned from data, and one is interested in achieving precise sound reasoning as opposed to approximate reasoning. We discuss each option briefly next.

In Option 1, it is desirable still to produce a symbolic description of the network for the sake of improving explainability (discussed further in Sect. 6.4) or trust, or for the purpose of communication and interaction with the system. In Option 2, by definition, a neurosymbolic interface is needed. This may be the best option currently, in practical terms, to combine reasoning and learning in AI given the apparent different nature of both tasks (the discrete and exact nature of logical reasoning and the continuous and approximate nature of statistical learning). However, the value of distributed approximate reasoning using neural networks is only starting to be explored as in the case of differentiable neural computers and neural theorem proving (Minervini et al. 2020). In Option 3, a reasonable requirement nowadays would be to compare results with deep learning and the other options. This is warranted by the latest practical results of deep learning showing that neural networks can offer, at least from a computational perspective, better results than purely symbolic systems.

Finally, the choice between Options 1 and 2 above may depend on the application at hand and the availability of quality data and knowledge. Scientists will continue to seek to make sense of the strengths and limitations of both neural and symbolic approaches. On this front, the research advances faster on the symbolic side due to the clear hierarchy of semantics and language expressiveness and rigour that exists at the foundation of the area. By contrast, little is known about the expressiveness of the latest DL models in relation to established neural models beyond data-driven comparative empirical evaluations. As advocated by Paul Smolensky, neurosymbolic computing can help map the latest neural models into existing symbolic hierarchies, thus helping organise the extensively ad-hoc body of work in neural computation (Huang et al. 2017).

7 Challenges for the principled combination of reasoning and learning

For a combined perspective on reasoning and learning, it is useful to note that reasoning systems may have difficulties computationally when reasoning with existential quantifiers and function symbols, such as $\exists x P(f(x))$. Efficient logic-based programming languages such as Prolog, for example, assume that every logical statement is universally quantified. By contrast, learning systems may have difficulty when adopting universal quantification over variables. To be able to learn a universally quantified statement such as $\forall x P(x)$, a learning systems needs in theory to be exposed to all possible instances of x. This simple

duality points to a possible complementary nature of the strengths of learning and reasoning systems. To learn efficiently $\forall x P(x)$, a learning system needs to *jump to conclusions*, extrapolating $\forall x P(x)$ given an adequate amount of evidence (the number of examples or instances of x). Such conclusions may obviously need to be revised over time in the presence of new evidence, as in the case of nonmonotonic logics. In this case, a statement of the form $\forall x P(x)$ becomes a data-dependent generalization, which is not to be assumed equivalent to a statement $\forall y P(y)$, as done in classical logic. Such statements may have been learned from different samples of the overall potentially infinite population. On the other hand, a statement of the form $\exists x P(x)$ is trivial to learn from data by identifying at least one case $P(a)$, although reasoning from $\exists x P(x)$ is more involved, requiring the adoption of an arbitrary constant b such that $P(b)$ holds.

It is now accepted that learning takes place in a continuous search space of (sub)differentiable functions; reasoning takes place in general in a discrete space as in the case of goal-directed theorem proving. The most immediate way of benefiting from the combination of reasoning and learning, therefore, is to adopt a hybrid approach whereby a neural network takes care of the continuous search space (and learning of probabilities), while a symbolic system consisting of logical descriptions of the network uses discrete search to achieve extrapolation and goal-directed reasoning. As hinted already in this paper, a property of early deep learning may hold the key to the above hybrid perspective: modularity. In the original paper on deep learning (Hinton et al. 2006), a modular system is proposed consisting of a stack of restricted Boltzmann machines. As a result of modularity, the extraction of symbolic descriptions from each RBM is made considerably easier (Tran and d'Avila Garcez 2018). Each RBM learns a joint probability distribution while their symbolic descriptions reflect the result of learning without manipulating probabilities explicitly, thus avoiding the complexity of probabilistic inference found in symbolic AI.

We therefore do not advocate the adoption of *monoblock* networks with millions of parameters. Even though this may be how the human brain works, loss of modularity seems to be, at least at present and from a computational perspective, a price that is too high to pay. Modularity remains a fundamentally relevant property of any computing system. In this space, we notice the emergence of neurosymbolic systems where large networks are combined with a "logical layer" under the assumption that the network (potentially with millions of parameters) carries out some relevant processing at a lower level of abstraction, e.g., object detection, before it is worth incorporating any logical concept at a higher-level of abstraction. An example would be the use of large Convolutional Neural Networks to which a neurosymbolic network is added only to the fully-connected layers at the end, as illustrated in Giunchiglia et al. (2022).

One major way of driving advances in AI continues to be through challenging applications, be it language translation, computer games or protein folding competitions. Language understanding in the broadest sense of the term, including question-answering that requires commonsense reasoning, offers probably the most complete application area for neurosymbolic AI. As an example, consider this question and its commonsense answer from the COPA data set (Roemmele et al. 2011): It got dark outside. What happened as a result? (a) Snowflakes began to fall from the sky; (b) The moon became visible in the sky. Another very relevant application domain is planning, which requires learning and reasoning over time, as in this example adapted from Schlag and Schmidhuber (2018): Daniel picks up the milk; Daniel goes to the bedroom; Daniel places the milk on the table; Daniel goes to the bathroom. Where is the milk? Finally, an area where machine learning and knowledge representation and reasoning have complementary strengths is *knowledge engineering*, including knowledge-base completion and data-driven ontology learning. In

this application area, rich and large-scale symbolic representations already exist alongside data, including knowledge graphs which can be combined with neural networks such as graph neural networks (Lamb et al. 2020; Carvalho et al. 2022). In the case of language models, it is difficult to see how the trained network with its learned embeddings could be discarded. The embeddings provide a context for concepts alongside their properties, that is, a grounding into the real-world for the symbols. This grounding is not normally present in purely symbolic form, a notable exception being Pierre Levy's Information Economy Meta-language (IEML) (Lévy 2013). IEML is designed bottom-up to account for semantics within the syntax of the language. It should be studied in the context of neurosymbolic AI because it makes it possible to calculate distances between concepts in symbolic form. When the neural network is expected to live side-by-side with the symbolic representation, the symbolic representation may be useful for reasoning about the network's answers that goes beyond the reasoning capabilities of the network. In this case the main objective is to link-up the symbols emerging at different parts of the network with their grounding. Establishing this connection, normally via the use of knowledge graphs, has been a main area of ongoing research as described in Carvalho et al. (2022).

Our conclusion from the above discussion is that in neurosymbolic AI:

(i) Knowledge should be grounded onto vector representations for efficient learning from data based on message passing in neural networks as an efficient computational model.

(ii) Symbols should become available as a result of querying and knowledge extraction from trained networks, and offer a rich description language at a proper level of abstraction, enabling *infinite uses of finite means*, but also compositional reasoning at the symbolic level allowing for extrapolation beyond the data distribution.[4]

(iii) The combination of learning and reasoning should offer an alternative to the problem of combinatorial reasoning by learning to reduce the number of effective combinations, thus producing simpler symbolic descriptions as part of the neurosymbolic cycle, that is, the translation of neural networks into symbolic representations and vice-versa.

As another example, consider a Variational Autoencoder which learns in unsupervised fashion to maximise mutual information between pixel inputs and a latent code or some other embedding consisting of fewer relevant features than the number of pixels. Suppose that this neural network has learned to find regularities such as e.g., *when it sees features of type A, it also sees features of type B but not features of type C*. At this point, such regularities can be converted into symbols: $\forall x A(x) \rightarrow (\exists y B(y) \wedge \neg \exists z C(z))$. As a result of the use of variables x, y, z at the symbolic level, one can extrapolate the above regularity to any feature of type A, B or C. A symbolic description is also a constraint on the neurosymbolic cycle. It is generalised from data during learning and it certainly includes an ability to ask *what-if* questions (c.f. Sect. 6). Reasoning about what has been learned allows for extrapolation beyond the data distribution, and finally the symbolic description can serve as prior knowledge (as a constraint) for further learning in the presence of more data, which includes the case of knowledge-based transfer learning. Further training can now take place at the perceptual or sub-symbolic level, or at the conceptual or symbolic

[4] Although knowledge extraction from large networks may not be complete, it should be sound and measurable (d'Avila Garcez et al. 2001; White and d'Avila Garcez 2019). One of the most interesting areas of recent progress addresses the emergence of symbols and their meaning from trained neural networks (Ngan et al. 2022).

level. This is when having a distributed (sub-symbolic) and a localist (symbolic or sub-symbolic) representation becomes relevant. Assuming that probabilities are dealt with at the sub-symbolic distributed level (as in RBMs) and that the symbolic level is used for a more qualitative representation of uncertainty in the form of general rules with exceptions, we avoid the complications of having to deal simultaneously with discrete and continuous learning of rules and probabilities.

A common challenge that will persist, however, is embodied in the question: *how symbolic meaning emerges from large networks of neurons?*

Perhaps an important choice for neurosymbolic AI is the choice between combinatorial (exact) reasoning and commonsense, approximate reasoning. While learning is always approximate, reasoning can be approximate or exact. In a neurosymbolic system, it is possible to envisage the combination of efficient approximate reasoning (jumping to conclusions) with more deliberative and precise or normative symbolic reasoning (d'Avila Garcez et al. 2015; Valiant 2003). Conclusions may be revised through learning from new observations and via communication with the system through knowledge extraction and exact reasoning. One might expect commonsense to emerge as a result of this process of reasoning and learning, although the modelling and computing with commonsense knowledge continues to be a challenge.

From a practical perspective, a recipe for neurosymbolic AI might be: learning is carried out from data by neural networks which use gradient-descent optimization; efficient forms of propositional reasoning can also be carried out by the network, c.f. neural-symbolic cognitive reasoning (d'Avila Garcez et al. 2009); rich first-order logic reasoning and extrapolation needs to be done symbolically from descriptions extracted from the trained network; once symbolic meaning has emerged from the trained network, symbols can be manipulated easily and serve as constraints for further learning from data; this establishes a feasible neurosymbolic AI cycle for learning and reasoning.

We therefore set out three immediate challenges for neurosymbolic AI, each capable of spinning out multiple research strands which in our view may become area defining in the next decade:

Challenge 1: First-order logic and higher-order knowledge extraction from very large networks that is provably sound and yet efficient, explains the entire model and local network interactions and accounts for different levels of abstraction.

Challenge 2: Goal-directed commonsense and efficient combinatorial reasoning about what has been learned by a complex deep network trained on large amounts of multimodal data.

Challenge 3: Human-network communication as part of a multi-agent system that promotes communication and argumentation protocols between the user and an agent that can ask questions and check her understanding.[5]

[5] The recent developments in Large Language Models, including OpenAI's ChatGPT, among others is related to our point. Humans in the loop, conversing and symbolically interacting with AI systems is an avenue that received attention recently under the umbrella of reinforcement learning with human feedback (RLHF) (Zhu et al. 2023). One can classify RLHF under the taxonomy of Neurosymbolic systems, as described by Kautz (2022).

8 Summary and future directions

Ingredients of Neurosymbolic AI Narrow AI based on neural networks is already successful in practice with big data. There is obvious value in this as shown by the flourishing of the ML community. Here, however, we have been discussing the science of what constitutes the fundamental ingredients of an intelligent system (Valiant 2003). One such ingredient, current results show, is gradient-based optimization used by DL to handle large amounts of data, but other ingredients are needed.

At the AAAI-2020 fireside conversation with Yoshua Bengio, a question was asked about the beauty and value of abstract compact symbolic representations such as $F = ma$ or $e = mc^2$. Bengio's answer was to point out that these must have come out of someone's brain (Isaac Newton and Albert Einstein). As Stephen Muggleton noted at another debate, his goal is to shorten the wait for the next Newton or Einstein. Muggleton's bet is on the use of higher-order logic representations in the place of deep learning. With this example we seek to illustrate that among highly respected researchers, the choice of AI ingredients may vary widely, from the need for much more realistic models of the brain to the need for ever more sophisticated forms of abstract computation. With the neurosymbolic methodology as a third way, the goal is to develop neural network models with a symbolic interpretation, to learn representations neurally and make them available for use symbolically (as for example when an AI system is asked to explain itself). In this paper, we have argued for *modularity* as an important ingredient, allowing one to refer to large parts of the network by the composition of symbols and relations among those symbols. Having an adequate *language for describing knowledge encoded in such networks* is another important ingredient. We have argued for the use of first-order logic as this language, as a canonical form of representation, but also other forms of non-classical logic. Once a complex network can be described symbolically, ideally by an abstract and compact form as in $e = mc^2$, any style of deductive reasoning becomes possible. *Reasoning* is obviously another fundamental ingredient, either within or outside the network, exact or approximate.

In what concerns System 1 (S1) and System 2 (S2) (Kahneman 2011), Kahneman was clear when he stated that he prefers to use *implicit* versus *explicit* thinking and reasoning. He argued that S1 (the intuitive parallel system) is capable of understanding language, in contradiction with Bengio's account of S1 and S2 for deep learning. Kahneman also stated that S2 (as the sequential deliberative system) is most probably performing symbol manipulation as argued by Marcus (2020). Marcus's attempt to define what makes a system hybrid is important inasmuch as a main problem of DL is a lack of definition. By contrast, LeCun's recent attempt at defining DL, also presented at AAAI 2020, falls short of a formal definition. LeCun's definition fails to distinguish deep and shallow networks. Nevertheless, all attempts to create bridges between S1 and S2 are at this point useful given our state of lack of understanding of how the brain works. Attempts to create differentiable reasoning are useful, although we would require an important distinction be made, whether the purpose is to achieve brain-like systems or to create robust AI. It is possible that these two goals may soon lead into two quite separate research directions: those who seek to understand and model the brain and those who seek to achieve or improve AI. Maybe from this perspective the field is too broad and will require further specialization. Finally, symbolic meaning can serve to improve performance of S1. In other words, symbols which have been learned, derived or even invented can act as constraints on the large network and help improve

learning performance as part of a positive *cycle of learning and reasoning*. Constraint satisfaction as part of the interplay between learning and reasoning is therefore another ingredient.

With the above five ingredients of neurosymbolic AI having been specified—*gradient-based optimization, modularity, symbolic language, reasoning and constraint satisfaction*—the reader will not be surprised to know that there are many outstanding challenges for neurosymbolic AI. First, no agreement exists on the best way of achieving the above combination of language and structure, of knowledge acquired by agents acting in an environment and the corresponding reasoning that an agent must implement to achieve its goals. It is highly desirable, though, that the study of how to achieve the combination of (symbolic) language and (neural) structure be principled, in that both language and structure should be specified formally with proofs about their correspondence or lack thereof. As done in the case of Noam Chomsky's language hierarchy (Chomsky 1956), proofs are needed of the capability of different neural architectures at representing various logical languages. Proofs of correspondence have been shown between neural networks and propositional, nonmonotonic, modal, epistemic and temporal logic (d'Avila Garcez et al. 2009). Similar proofs are required for first-order and higher-order logic. The choice of an appropriate (logical or probabilistic) language for neurosymbolic AI will play a central role in the technical debate. In this process, we argue: translate back and forth between representations, take a principled approach, adopt a language as a constraint on the structure, seek to provide explanations, combine learning and reasoning, apply, revise and repeat.

Finally, neurosymbolic AI will need standard benchmarks and associated comprehensibility tests which could, in a principled way, offer a fair comparative evaluation with other approaches. With *benchmarks* in neurosymbolic AI, we are less interested in rankings and leaderboards (however useful these are), and we are interested more in the definition of suitable metrics and representative examples for neurosymbolic AI. Metrics that go beyond accuracy to address computational complexity, energy consumption and accountability in AI, and representative examples, relatively simple and clearly defined tasks, as in the case of the international SAT competition (http://www.satcompetition.org/), capable of addressing specific learning, knowledge representation and reasoning challenges. The kind of benchmark for the 3rd wave of AI should address issues related to: learning from fewer examples, accuracy w.r.t. computing power, performance in the case of extrapolation, transfer learning and concept learning, representative reasoning outside of the training distribution, e.g., reasoning with variables and reasoning about time, knowledge, uncertainty, etc. The design of such benchmarks has already started, see e.g., d'Avila Garcez and Jiménez-Ruiz (2022). Just as the field of AI progressed when challenging applications were set such as chess playing, robot football, self-driving vehicles and protein folding, neurosymbolic AI should benefit from similar challenges being set by the AI community specifically for it in the next decade.

Acknowledgements We would like to thank Gary Marcus, Francesca Rossi, Pascal Hitzler, Marco Gori, Kristian Kersting, Pierre Lévy, Luc de Raedt, Alessandra Russo, Kristian Kersting, Murray Shanahan, Alexander Gray, Moshe Vardi and the anonymous reviewers who helped us improve this manuscript. Luis Lamb is supported in part by CNPq and CAPES - Finance Code 001.

References

Arabshahi F, Lu Z, Singh S, Anandkumar A (2019) Memory augmented recursive neural networks. https://arxiv.org/abs/1911.01545

Bader S, Hitzler P, Hölldobler S, Witzel A (2007) The core method: Connectionist model generation for first-order logic programs. In: Hammer B, Hitzler P (eds) Perspectives of neural-symbolic integration. Springer, Berlin, pp 205–232. https://doi.org/10.1007/978-3-540-73954-8_9

Bengio Y, Deleu T, Rahaman N, Ke NR, Lachapelle S, Bilaniuk O, Goyal A, Pal CJ (2020) A meta-transfer objective for learning to disentangle causal mechanisms. In: ICLR

Cameron C, Chen R, Hartford J, Leyton-Brown K (2020) Predicting propositional satisfiability via end-to-end learning. In: AAAI

Carvalho BW, Garcez AD, Lamb LC (2022) Graph-based neural modules to inspect attention-based architectures: a position paper. In AAAI fall symposium 2022. Arlington, Virginia. https://doi.org/10.48550/ARXIV.2210.07117. https://arxiv.org/abs/2210.07117

Chaudhuri S, Ellis K, Polozov O, Singh R, Solar-Lezama A, Yue Y (2021) Neurosymbolic programming. Found Trends Program Lang 7(3):158–243

Chen J, Batmanghelich K (2019) Weakly supervised disentanglement by pairwise similarities. https://arxiv.org/abs/1906.01044

Chomsky N (1956) Three models for the description of language. IRE Trans Inf Theory 2(3):113–124. https://doi.org/10.1109/TIT.1956.1056813

d'Avila Garcez AS, Besold T, de Raedt L, Földiák P, Hitzler P, Icard T, Kühnberger K, Lamb LC, Miikkulainen R, Silver D (2015) Neural-symbolic learning and reasoning: Contributions and challenges. In: AAAI Spring Symposia. http://www.aaai.org/ocs/index.php/SSS/SSS15/paper/view/10281

d'Avila Garcez AS, Jiménez-Ruiz E (eds) (2022) Proceedings of the 16th international workshop on neural-symbolic learning and reasoning. Cumberland Lodge, Windsor Great Park, Sept 28–30. CEUR workshop proceedings, vol 3212. CEUR-WS.org, Aachen. http://ceur-ws.org/Vol-3212

d'Avila Garcez AS, Lamb LC (2003) Reasoning about time and knowledge in neural symbolic learning systems. In: NIPS, pp 921–928

d'Avila Garcez AS, Zaverucha G (1999) The connectionist inductive learning and logic programming system. Appl Intell 11(1):59–77. https://doi.org/10.1023/A:1008328630915

d'Avila Garcez AS, Broda K, Gabbay DM (2001) Symbolic knowledge extraction from trained neural networks: a sound approach. Artif Intell 125(1–2):155–207. https://doi.org/10.1016/S0004-3702(00)00077-1

d'Avila Garcez A, Broda K, Gabbay DM (2002) Neural-symbolic learning systems: foundations and applications. Springer, Berlin

d'Avila Garcez AS, Lamb LC, Gabbay DM (2009) Neural-symbolic cognitive reasoning. Springer, Berlin-Heidelberg

d'Avila Garcez AS, Gori M, Lamb LC, Serafini L, Spranger M, Tran S (2019) Neural-symbolic computing: an effective methodology for principled integration of machine learning and reasoning. FLAP 6(4):611–632

De Raedt L, Kimmig A, Toivonen H (2007) Problog: a probabilistic prolog and its application in link discovery. In: Proceedings of the 20th international joint conference on artifical intelligence. IJCAI'07, pp 2468–2473. Morgan Kaufmann Publishers Inc., San Francisco. http://dl.acm.org/citation.cfm?id=1625275.1625673

Evans R, Grefenstette E (2018) Learning explanatory rules from noisy data. JAIR 61:1–64

Fagin R, Halpern JY, Moses Y, Vardi MY (2003) Reasoning about knowledge. MIT Press, Cambridge

Giunchiglia E, Stoian MC, Lukasiewicz T (2022) Deep learning with logical constraints. In: IJCAI-ECAI 2022. IJCAI/AAAI Press, Vienna

Gori M (2018) Machine learning: a constraint-based approach. Morgan Kaufmann, Burlington

Hammer B, Hitzler P (eds) (2007) Perspectives of neural-symbolic integration. Springer, Berlin

Hinton GE, Osindero S, Teh Y (2006) A fast learning algorithm for deep belief nets. Neural Comput 18(7):1527–1554. https://doi.org/10.1162/neco.2006.18.7.1527

Hochreiter S (2022) Toward a broad AI. Commun ACM 65(4):56–57

Huang Q, Smolensky P, He X, Deng L, Wu D (2017) A neural-symbolic approach to natural language tasks. https://arxiv.org/abs/1710.11475

Hu Z, Ma X, Liu Z, Hovy E, Xing E (2016) Harnessing deep neural networks with logic rules. In: ACL

Jumper J, Evans R, et al (2020) High accuracy protein structure prediction using deep learning. In: 14th critical assessment of techniques for protein structure prediction, CASP-14

Kahneman D (2011) Thinking. Fast and slow. Farrar, Straus and Giroux, New York

Kautz HA (2022) The third AI summer: AAAI Robert S. Engelmore memorial lecture. AI Mag 43(1):105–125. https://doi.org/10.1002/aaai.12036

Kleinberg JM, Ludwig J, Mullainathan S, Sunstein CR (2020) Algorithms as discrimination detectors. Proc Natl Acad Sci USA 117(48):30096–30100. https://doi.org/10.1073/pnas.1912790117

Kosiorek AR, Sabour S, Teh YW, Hinton GE (2019) Stacked capsule autoencoders. In: NeurIPS 2019, pp 15486–15496. http://papers.nips.cc/paper/9684-stacked-capsule-autoencoders
Lamb LC, d'Avila Garcez AS, Gori M, Prates M, Avelar P, Vardi MY (2020) Graph, neural networks meet neural-symbolic computing: a survey and perspective. IJCAI 2020:4877–4884
Lample G, Charton F (2020) Deep learning for symbolic mathematics. In: ICLR. https://openreview.net/forum?id=S1eZYeHFDS
LeCun Y, Bengio Y, Hinton G (2015) Deep learning. Nature 521(7553):436–444
Lévy P (2013) The semantic sphere 1: computation, cognition and information economy. ISTE. Wiley, Hoboken. https://books.google.co.uk/books?id=EIhS9DqwLkgC
Lloyd JW (2003) Logic for learning - learning comprehensible theories from structured data. Cognitive Technologies. Springer, Berlin
Manhaeve R, Dumancic S, Kimmig A, Demeester T, De Raedt L (2018) Deepproblog: neural probabilistic logic programming. In: Bengio S, Wallach H, Larochelle H, Grauman K, Cesa-Bianchi N, Garnett R (eds) NIPS 31, Montreal, pp 3749–3759
Mao J, Gan C, Kohli P, Tenenbaum J, Wu J (2019) The neuro-symbolic concept learner: interpreting scenes, words, and sentences from natural supervision. In: ICLR
Marcus G (2020) The next decade in AI: four steps towards robust artificial intelligence. https://arxiv.org/abs/1801.00631
Marra G, Diligenti M, Giannini F, Gori M, Maggini M (2020) Relational neural machines. https://arxiv.org/abs/2002.02193
Marra G, Giannini F, Diligenti M, Gori M (2019) Lyrics: a general interface layer to integrate logic inference and deep learning. In: Joint European conference on machine learning and knowledge discovery in databases, Springer, pp 283–298
McCarthy J (1988) Epistemological challenges for connectionism. Behav Brain Sci 11(1):44–44. https://doi.org/10.1017/S0140525X0005264X
Minervini P, Bosnjak M, Rocktäschel T, Riedel S, Grefenstette E (2020) Differentiable reasoning on large knowledge bases and natural language. In: AAAI, pp 5182–5190
Ngan KH, Garcez AD, Townsend J (2022) Extracting meaningful high-fidelity knowledge from convolutional neural networks. In: 2022 international joint conference on neural networks (IJCNN), pp 1–17 . https://doi.org/10.1109/IJCNN55064.2022.9892194
Page M (2000) Connectionist modelling in psychology: a localist manifesto. Behav Brain Sci 23(4):443–467. https://doi.org/10.1017/S0140525X00003356
Pearl J (2019) The seven tools of causal inference, with reflections on machine learning. Commun ACM 62(3):54–60. https://doi.org/10.1145/3241036
Prates MOR, Avelar PHC, Lemos H, Lamb LC, Vardi MY (2019) Learning to solve NP-complete problems: a graph neural network for decision TSP. In: AAAI
Raedt LD, Kersting K, Natarajan S, Poole D (2016) Statistical relational artificial intelligence: logic, probability, and computation. Synthesis lectures on artificial intelligence and machine learning. Morgan & Claypool, San Rafael
Richardson M, Domingos P (2006) Markov logic networks. Mach Learn 62(1–2):107–136
Riegel R, Gray A, Luus F, Khan N, Makondo N, Akhalwaya I, Qian H, Fagin R, Barahona F, Sharma U, Ikbal S, Karanam H, Neelam S, Likhyani A, Srivastava S (2020) Logical neural networks. https://arxiv.org/abs/2006.13155
Rocktäschel T, Riedel S (2017) End-to-end differentiable proving. https://arxiv.org/abs/1705.11040
Roemmele M, Bejan CA, Gordon AS (2011) Choice of plausible alternatives: an evaluation of commonsense causal reasoning. In: AAAI spring symposia. Stanford
Schlag I, Schmidhuber J (2018) Learning to reason with third-order tensor products. In: NeurIPS
Serafini L, d'Avila Garcez AS (2016) Logic tensor networks: deep learning and logical reasoning from data and knowledge. https://arxiv.org/abs/1606.04422
Socher R, Chen D, Manning C, Ng A (2013) Reasoning with neural tensor networks for knowledge base completion. In: NIPS, pp 926–934
Stehr M-O, Kim M, Talcott CL (2022) A probabilistic approximate logic for neuro-symbolic learning and reasoning. J Log Algebr Methods Program 124:100719. https://doi.org/10.1016/j.jlamp.2021.100719
Tavares AR, Avelar PHC, Flach JM, Nicolau M, Lamb LC, Vardi MY (2020) Understanding boolean function learnability on deep neural networks. https://arxiv.org/abs/2009.05908
Tran S, d'Avila Garcez A (2018) Deep logic networks: inserting and extracting knowledge from deep belief networks. IEEE TNNLS 29:246–258. https://doi.org/10.1109/TNNLS.2016.2603784
Valiant LG (2003) Three problems in computer science. J ACM 50(1):96–99
White A, d'Avila Garcez A (2019) Measurable counterfactual local explanations for any classifier. https://arxiv.org/abs/1908.03020

Xu J, Zhang Z, Friedman T, Liang Y, den Broeck GV (2018) A semantic loss function for deep learning with symbolic knowledge. In: Dy JG, Krause A (eds) ICML. Proceedings of machine learning research, vol 80, Stockholm

Yang F, Yang Z, Cohen WW (2017) Differentiable learning of logical rules for knowledge base reasoning. In: NIPS, pp 2319–2328. http://papers.nips.cc/paper/6826-differentiable-learning-of-logical-rules-for-knowledge-base-reasoning.pdf

Zhu B, Jiao J, Jordan MI (2023) Principled reinforcement learning with human feedback from pairwise or K-wise comparisons. https://doi.org/10.48550/ARXIV.2301.11270. https://arxiv.org/abs/2301.11270

Publisher's Note Springer Nature remains neutral with regard to jurisdictional claims in published maps and institutional affiliations.

Springer Nature or its licensor (e.g. a society or other partner) holds exclusive rights to this article under a publishing agreement with the author(s) or other rightsholder(s); author self-archiving of the accepted manuscript version of this article is solely governed by the terms of such publishing agreement and applicable law.